

Loss Functions

Moin Mostakim

Department of Computer Science and Engineering
Faculty of School of Data Science

October 2023



Contents of the slide

- ① Mean Squared Error (MSE)
- ② Mean Absolute Error (MAE)
- ③ Huber Loss
- ④ Binary Cross-Entropy Loss
- ⑤ Categorical Cross-Entropy Loss
- ⑥ Sparse Categorical Cross-Entropy Loss
- ⑦ Hinge Loss
- ⑧ Kullback-Leibler Divergence (KL Divergence)

Loss Functions in Machine Learning

- In machine learning, loss functions are crucial for training models. They quantify the error between predicted and actual values.
- Choosing the right loss function depends on the task at hand. Here are a few common loss functions and their use cases.

Mean Squared Error (MSE)

Task: Regression

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1)$$

where n is the number of samples, y_i is the true target, and $\hat{y}_i$ is the predicted value.

Mean Absolute Error (MAE)

Task: Regression

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

Task: Regression

$$Huber(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{if } |y - \hat{y}| \leq \delta \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \quad (3)$$

where δ is a hyperparameter that determines the threshold for the transition from quadratic to linear behavior.

Binary Cross-Entropy Loss (Log Loss)

Task: Binary Classification

$$BCE(y, \hat{y}) = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})) \quad (4)$$

where y is the true label (0 or 1), and $\hat{y}$ is the predicted probability of class 1.

Categorical Cross-Entropy Loss

Task: Multiclass Classification

$$CCE(y, \hat{y}) = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (5)$$

where N is the number of classes, y_i is the true distribution over classes (one-hot encoded), and $\hat{y}_i$ is the predicted probability distribution.

Sparse Categorical Cross-Entropy Loss

Task: Multiclass Classification with integer class labels

$$SCCE(y, \hat{y}) = -\log(\hat{y}_y) \quad (6)$$

where y is the true class label, and $\hat{y}_y$ is the predicted probability for the true class.

Hinge Loss (SVM Loss)

Task: Support Vector Machines (SVM)

$$\text{Hinge}(y, \hat{y}) = \max(0, 1 - y \cdot \hat{y}) \quad (7)$$

where y is the true class label (-1 or 1), and $\hat{y}$ is the predicted score.

Kullback-Leibler Divergence (KL Divergence)

Task: Measuring dissimilarity between probability distributions

$$KL(p||q) = \sum_i p(i) \log \left(\frac{p(i)}{q(i)} \right) \quad (8)$$

where p and q are the true and predicted probability distributions, respectively.

Task: Siamese networks for similarity learning

$$TL(A, P, N) = \max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0) \quad (9)$$

where A is the anchor sample, P is a positive sample, N is a negative sample, f is the embedding function, and α is a margin that separates positive and negative pairs.